

Resonance structures (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- Deduce the resonance structure of molecules and ions.
- Outline the role of electron delocalisation in formation of resonance hybrid structures.

The Earth's atmosphere is composed of many different layers based on unique properties such as temperature and pressure. The layer closest to Earth is known as the troposphere and many of the gases that make it up – nitrogen, oxygen, carbon dioxide, methane, water vapour – are molecules that you have already drawn Lewis formulas for.

The atmospheric layer above this, the stratosphere, has an entirely different composition. 90% of the trioxxygen found on Earth, more commonly known as ozone O_3 , exists in the stratosphere (**Figure 1**).

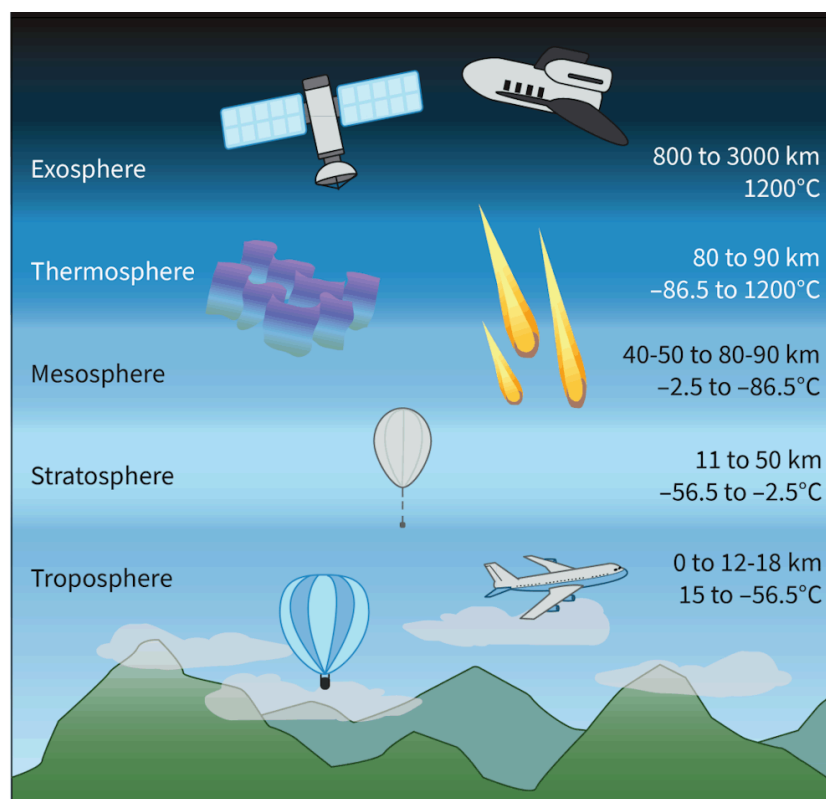


Figure 1. Earth's atmosphere.

 More information for figure 1

The image is a diagram illustrating Earth's atmospheric layers from the ground upwards. At the bottom is the Troposphere, ranging from 0 to 12-18 km in altitude, with temperatures from 15 to -56.5°C . Above this is the Stratosphere, 11 to 50 km high, with temperatures between -56.5 and -2.5°C . The Mesosphere comes next, from 40-50 to 80-90 km, with temperatures ranging from -2.5 to -86.5°C . The Thermosphere is above the Mesosphere, spanning 80 to 90 km with temperatures from -86.5 to 1200°C . Finally, the Exosphere extends from 800 to 3000 km with a temperature of 1200°C . Each layer is visually represented with different objects: a mountain and clouds for the Troposphere, an airplane for the Stratosphere, balloons for upper layers, fiery meteors for the Mesosphere, and a satellite and spaceship in the higher Thermosphere and Exosphere. The layers are represented with labels indicating their names, altitude range, and temperature range.

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Analysis of the ozone molecule reveals that the bonds in it can be broken by light with a wavelength of 330 nm. This wavelength is longer than the wavelength required to break the double bond in an oxygen molecule, 242 nm.

Interestingly, the wavelength required to break the bond in ozone is shorter than the wavelength that would be required to break an oxygen–oxygen single bond, 845 nm (**Table 1**).

Table 1. Wavelength required to break bonds.

Bond	O–O	Bonds in ozone
Wavelength of light required to break the bond (nm)	845	330

How is the wavelength of light required to break the bond related to the amount of energy required? What does this suggest about the ozone bond? How do you think the bond lengths for ozone would compare in relation to the oxygen–oxygen single bonds and double bonds?

Making connections

In [section S2.2.7 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/covalent-network-structures-id-43593/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/covalent-network-structures-id-43593/) the term allotrope was defined for you as a different physical form of the same element. Ozone, IUPAC name trioxygen, is an allotrope of the much more common form dioxygen, oxygen.

Resonance

[Section S2.2.2 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/multiple-covalent-bonds-id-45128/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/multiple-covalent-bonds-id-45128/) covered the steps for drawing Lewis formulas. When asked to draw the Lewis formula for Ozone, O_3 , one student drew the formula shown in **Figure 2**:

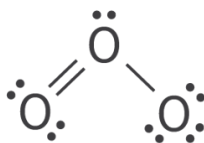


Figure 2. Student A's Lewis formula of ozone.

 More information for figure 2

The image represents a Lewis structure of an ozone (O_3) molecule. It consists of three oxygen atoms arranged in a bent shape, with one oxygen atom double bonded to the central oxygen, and the other single bonded. Each oxygen atom shows its valence electrons: the central atom has one lone pair, while the outer atoms have three lone pairs each.

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Another student drew the structure in **Figure 3** instead:

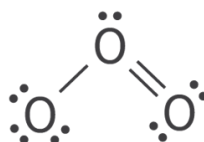


Figure 3. Student B's Lewis formula of ozone.

 More information for figure 3

The image shows a Lewis structure representation of an ozone (O_3) molecule. It consists of three oxygen atoms arranged in a bent or V-shaped configuration. The central oxygen atom is connected to the two terminal oxygen atoms via single and double bonds. The central oxygen atom has two lone pairs of electrons at the top. Each of the terminal oxygen atoms has two lone pairs of electrons on two opposite sides. This arrangement suggests a resonance structure in which the electrons are delocalized across the entire molecule.

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Compare these two Lewis formulas. Note that both have three pairs of bonding electrons, a V-shaped or bent molecular geometry and a bond angle of 117° , but these structures are not the same. Which student is correct?

According to Lewis theory, both are valid: both structures show the 18 valence electrons, which O_3 would have, and all atoms of oxygen in both the structures satisfy the octet rule.

It is through analysis of the experimental evidence discussed at the beginning of this section that the true structure came to be realised. This true structure, known as the resonance structure, is a hybrid of these two Lewis formulas drawn by the students, as shown in **Figure 4**. Neither of these actually show the true structure, the hybrid is understood to be an intermediate structure. The double-headed arrow, $\longleftrightarrow$, in **Figure 4** shows that this is a resonance structure.



Figure 4. The resonance structure of ozone.

More information for figure 4

The image is a diagram illustrating the resonance structures of ozone. It displays two ozone molecules, represented as Lewis structures, with a double-headed arrow ($\leftrightarrow$) between them to indicate resonance. Each molecule consists of three oxygen atoms, with the left-hand side structure showing a double bond between the first and second oxygen atoms and a single bond between the second and third. The right-hand side structure displays a single bond between the first and second oxygen atoms and a double bond between the second and third. Each oxygen atom is surrounded by pairs of dots representing lone pair electrons.

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The bonding in an ozone molecule is not one single O–O and one double O=O bond, but two identical bonds that are something in between. The two single bonds remain unchanged, but the pair of electrons in the ozone molecule that would be used to form a double bond are instead spread evenly over the two bonding positions. The molecule achieves this by delocalising one of the pairs of electrons. That means it is no longer fixed to one bonding position: it is spread over two positions.

This electron delocalisation is often represented using a dashed line, as shown in **Figure 5**. Note that **Figure 5** is not a Lewis formula as it does not show the non-bonding electrons.

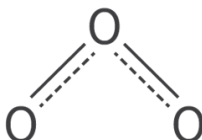


Figure 5. The resonance structure of ozone represented with dashed lines.

More information for figure 5

The image is a diagram representing the resonance structure of an ozone molecule. It consists of three oxygen atoms connected in a bent arrangement. The central oxygen atom is bonded to the two outer oxygen atoms. The bonds between the oxygen atoms are depicted using dashed lines, indicating that the electron pairs are delocalized and spread over both bonding positions. This depiction shows that the bonds are equivalent and share characteristics of both single and double bonds, a key feature of resonance structures. The diagram visually represents the concept that the electrons are not fixed in place but are distributed across the structure, illustrating electron delocalization.

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Resonance hybrid structures can be used to represent molecules for which there are more than one valid Lewis formula in terms of electron distribution. Note that the placement of the atoms has not changed, this is a key idea we will revisit later.

Nature of Science

Aspect: Theories

The fact that two different Lewis formulas, both technically correct, can be drawn for ozone highlights some of the limitations of the Lewis theory and importance of experimental evidence in deducing the molecular structure.

Resonance hybrid structures are so called because the bonds have hybrid or intermediate properties (**Figure 6**).

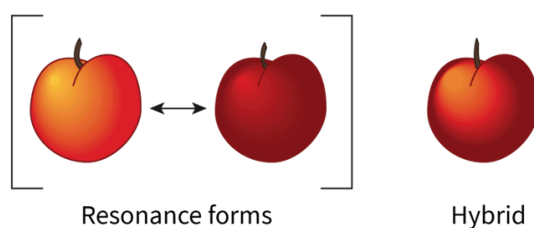


Figure 6. Resonance hybrid structures have intermediate properties.

 More information for figure 6

The image depicts two illustrations of apples labeled as "Resonance forms" on the left with a double-headed arrow between them, symbolizing a transformation or transition. On the right, there's a single illustration labeled "Hybrid" indicating a combination or intermediary state. The apples visually represent the concept of resonance structures transitioning into a hybrid state, akin to chemical structures that have intermediate properties. This visual is an analogy used to explain the concept of resonance in chemistry, where individual structures contribute to a hybrid with unique characteristics.

[Generated by AI]

In the initial Lewis formula shown, we would expect that ozone would have two different bond lengths and bond strengths. Experimental evidence shows that the bonds in ozone have intermediate properties between singly and doubly bonded oxygen. This concept of resonance hybrids can be used to explain the intermediate bond strengths described previously: the bonds are stronger than a single bond but weaker than a double bond. This is why ozone molecules require a longer wavelength of light to break their bonds than the $\text{O}=\text{O}$ double bond in oxygen molecules, but a shorter wavelength than what would be needed for an oxygen–oxygen single bond.

How can this be used to explain the data for bond length given in **Table 2**?

Table 2. Comparative bond lengths.

Bond	O—O	Bonds in ozone	C
Bond length (nm)	0.148	0.128	C

Study skills

A common area of confusion for students is to think that the hybrid structures are constantly switching between the two forms, that is, that the double bond is switching positions, sometimes in one location and sometimes in another.

This is not the case! In reality, the second pair of electrons that would be responsible for making the bond 'double' are being shared over several locations and are spread evenly between the bonding positions.

Think of it as being similar to a mule, which is a hybrid of a donkey and a horse — it is not a donkey one minute and a horse the next!

Bond order

Think back to the properties of single, double and triple bonds covered in [section S2.2.2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/multiple-covalent-bonds-id-45128/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/multiple-covalent-bonds-id-45128/>). They differed in terms of bond length (single bonds are the longest bonds) and bond strength (single bonds are the weakest bonds). Collectively, these properties can be referred to as bond order. Single bonds have a bond order of 1, double bonds have a bond order of 2 and the bond order of a triple bond is 3.

In a resonance hybrid such as ozone, the bond order is 1.5 or $\frac{3}{2}$. The single bond between each oxygen has a bond order of 1 and the pair of electrons which are spread over 2 bonding positions is responsible for the $\frac{1}{2}$.

Other molecules and ions also show resonance

Any molecule or ion whose electron distribution can be represented by more than one valid Lewis formula will demonstrate resonance. Take the carbonate ion, CO_3^{2-} as an example (**Figure 7**). The total number of valence electrons in this ion is 24 and so all three Lewis formulas are valid. What feature of the ion is responsible for its ability to form resonance structures?

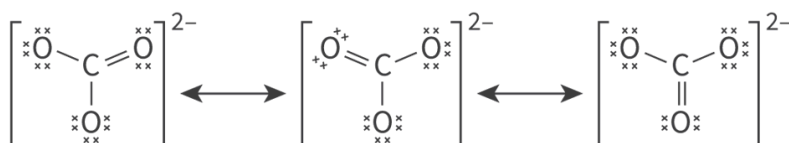


Figure 7. Resonance hybrid structures for the carbonate ion.

 More information for figure 7

The image shows the resonance hybrid structures for the carbonate ion, CO_3^{2-} . There are three distinct Lewis structures displayed clearly. Each structure consists of a carbon atom centrally located and bonded to three oxygen atoms. In each structure, one of the oxygen atoms forms a double bond with the carbon, while the other two oxygen atoms form single bonds, resulting in a total charge of -2. The structures are connected using double-

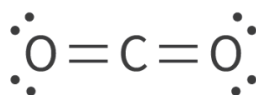
headed arrows indicating resonance. The double bonds are shown in different positions: on the left, center, and right of the carbon in each respective structure. This illustrates the concept of resonance where the placement of the double bond varies among the three oxygen atoms.

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In the carbonate ion, there are three possible places for the double bond to be placed. Having this 'choice' of where to place the double bond is why an ion or molecule will demonstrate resonance. As there are three possible places for the double bond to be placed, a pair of electrons is spread evenly over the three options. This results in a bond order of 1.333 or $\frac{4}{3}$.

Let's analyse two other species, a carbon dioxide molecule CO_2 , and the carbonite ion, CO_2^{2-} . Both species are represented using Lewis formulas in **Figure 8**. Which of these species will demonstrate resonance?

Carbon dioxide:



Carbonite ion:

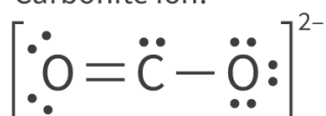


Figure 8. Lewis formulas of carbon dioxide molecule and carbonite ion.

 More information for figure 8

The image displays Lewis structures for two chemical species. The first structure represents the carbon dioxide molecule: it consists of a central carbon atom (C) double-bonded to two oxygen atoms (O), one on each side, portrayed as $\text{O}=\text{C}=\text{O}$. Dots around the oxygen atoms show the valence electrons, with four dots on each oxygen.

The second structure represents the carbonite ion with two oxygen atoms flanking a central carbon atom, depicted as $[\text{O}=\text{C}-\text{O}]^{2-}$. This structure also uses dots to indicate valence electrons, with four dots around each oxygen and two dots above the carbon, indicating a negative charge for the ion. This structure is inside square brackets, and the charge is shown as 2- outside the brackets, on the upper right corner.

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As there is no 'choice' of where to place the $\text{C}=\text{O}$ bond in the carbon dioxide molecule, this cannot experience resonance. In the carbonite ion, there are two possible places for the $\text{C}=\text{O}$ bond; this ion can experience resonance (**Figure 9**).

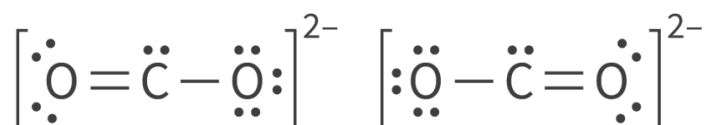
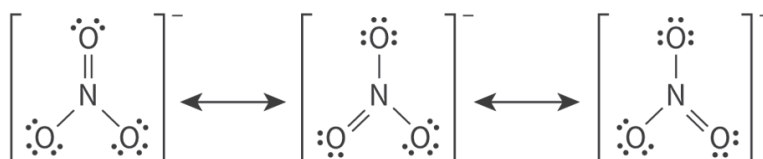


Figure 9. Resonance hybrid structures of carbonite ion.
 More information for figure 9

The image displays two resonance structures for the carbonite ion. Both structures are encased in square brackets with a 2- charge indicated outside the upper right corner of the brackets. In the first structure, from left to right, an oxygen atom with a double bond connects to a carbon atom, which is single-bonded to another oxygen. The left oxygen atom has two lone pairs, the carbon atom is depicted with no lone pairs, and the right oxygen atom with three lone pairs. In the second structure, the positions of the double and single bonds switch. The left oxygen atom is single-bonded and has three lone pairs, the carbon atom has no lone pairs, and the right oxygen atom is double-bonded with two lone pairs.

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To determine the valid resonance structures, first draw the Lewis formula using the steps in [section S2.2.2 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/multiple-covalent-bonds-id-45128/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/multiple-covalent-bonds-id-45128/). Then, count the number of potential positions for the double bond. For example, the nitrate ion, NO_3^- , has three N–O bonds over which an extra bonding pair of electrons are to be placed, giving three different potential positions for the double bond to be positioned. Therefore, there are three resonance structures for the nitrate ion (**Figure 10**). What would be the bond order of the nitrate ion? Because the nitrogen atom has four bonding pairs of electrons spread over three bonding positions, the bond order would be $\frac{4}{3}$, therefore $1\frac{1}{3}$ bond.

**Figure 10.** Resonance hybrid structures of the nitrate ion.
 More information for figure 10

The image shows three resonance structures of the nitrate ion (NO_3^-). Each structure is enclosed in brackets and includes a negative charge outside the brackets. In all three structures, a nitrogen atom is centrally bonded to three oxygen atoms. The structures vary by the position of the double bond:

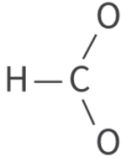
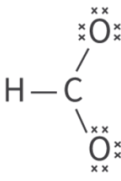
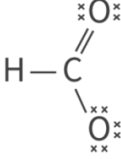
1. The first structure has a double bond between the nitrogen and the top oxygen, with single bonds to the other two oxygens. Each oxygen atom not involved in the double bond has three lone pairs of electrons.
2. The second structure features a double bond with the oxygen on the left, while the nitrogen forms single bonds with the other two oxygens. Again, lone pairs are present on each singly bonded oxygen.
3. The third structure shows the double bond with the right oxygen, with the other two oxygens singly bonded to nitrogen, and lone pairs on the singly bonded oxygens.

Arrows between the structures indicate the resonance form, emphasizing that the actual nitrate ion is a hybrid of these three structures. The presence of different resonance structures shows the delocalization of electrons in the nitrate ion, contributing to its stability.

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Worked example 1

Draw the possible resonance structures for the methanoate ion, HCOO^- .

Solution steps	Calculations
Step 1: Calculate total valence electrons,	$\text{H} + \text{C} + (\text{O} \times 2) + (\text{negative charge}) =$
Step 2: Identify the central atom. Here it will be carbon as it is further left on the periodic table than oxygen, it is the least numerous (only 1 carbon compared to 2 oxygens) and hydrogen cannot be a central atom. Join all peripheral atoms (hydrogen and oxygens) to the central atom (carbon) with single bonds.	
Step 3: Add remaining electrons as valence electrons,	
Step 4: Add double or triple bonds where necessary to ensure that all atoms fulfil octet rule (<i>carbon was not fulfilling the rule otherwise</i>).	

Solution steps	Calculations
Step 5: Check number of electrons added is as calculated in Step 1.	18 electrons are added in total. As this is an ion, Check whether there are alternative positions for electrons making up the double bond can be placed and the carbon: this means that the ion will show If the species shows resonance, draw all possible structures with arrows between them.

Study skills

When asked to draw the Lewis formula for the resonance hybrid structures, you must draw all the potential Lewis formulas as shown in Worked example 1, including the non-bonding pairs.

In this activity, you will have a chance to apply your knowledge of resonance structures to different ions present in substances that are used in everyday life.

Activity

- **IB learner profile attribute:**
 - Knowledgeable

◦ Thinker

- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual activity

1. Toothpastes containing sodium hydrogen carbonate, HCO_3^- , are widely used. Draw Lewis formulas for the hydrogen carbonate ion, with resonance forms where appropriate.
2. Nitrites are widely used in food preservation. The nitrite ion, NO_2^- displays resonance. Draw each of these resonance structures (there are two). Predict the bond length and bond strength for the nitrogen—oxygen bond. Then, research the real value to see how close your estimate was.